Pi 5 EM Covert Channel — “SOS” Sent and Recovered

Phase 1 payload recovery | 2026-11-10 | R. Boada

Milestone: the Raspberry Pi 5 transmitted the message “SOS” in Morse code purely through its electromagnetic emissions, and we recovered it over the air on a HackRF software-defined radio — with zero errors. This completes the Phase 1 goal: show the board's memory-bus activity can carry a chosen data payload off the device, not just leak that it is busy.

How it works

- Transmitter (Pi, no special hardware). tx_ddr --morse SOS --unit 250 keys the Pi's 286.8 MHz memory-bus emission on and off in standard Morse timing: streaming writes to DRAM = a “mark” (carrier on), an idle pause = a “space” (carrier off). The data is the on/off pattern.

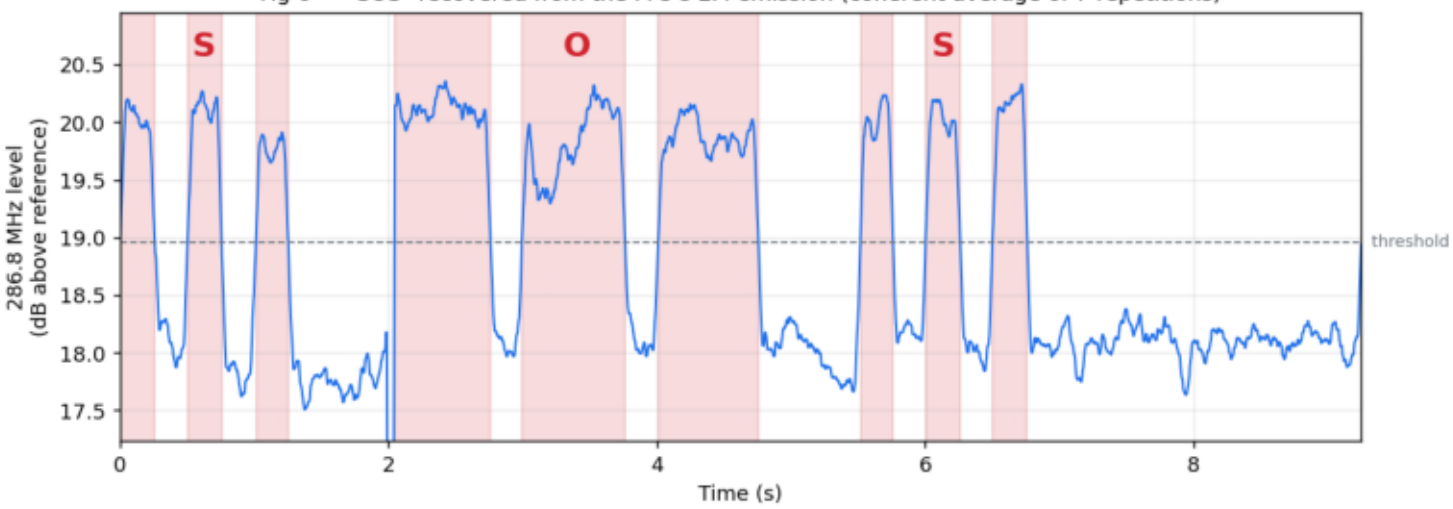
- Receiver (HackRF One @ 286.8 MHz). Driven directly through libhackrf to work around the board's old 2015 firmware. The tinySA can detect the emission but is too slow to decode keying; the HackRF provides the continuous IQ stream a decoder needs.

- The catch. Each Morse symbol changes the received level by only ~2 dB — below the ambient RF fluctuation — so a single pass cannot be decoded cleanly.

- The fix: coherent averaging (the Phase 1 trick). Because the message repeats, we capture ~70 s, find the repeat period by autocorrelation, align each repetition to cancel timing jitter, and fold them together. The periodic keying reinforces while ambient noise averages down. The averaged envelope (below) then shows a clean S–O–S, decoded with 0 bit errors.

- Honest scope. Short range (near-field, tens of cm with a whip), and recovery relies on averaging the repeated message — exactly the methodology used to find the emission in the first place. A near-field probe or a quieter enclosure would extend this; so would encoding without repetition.

Fig 6 — "SOS" recovered from the Pi 5's EM emission (coherent average of 7 repetitions)



Method: HackRF @ 286.8 MHz, 2 MS/s; band-power envelope (line peak minus a reference band); autocorrelation period estimate + per-repetition aligned fold; 2-means threshold + Morse run-length decode.